40% chance of living beyond age eighty-five, and females have more than a 50% chance. For healthy people, these probabilities increase to 50% for males and 62% for females.

The beneficiaries of Rhode Island's pension plan do not face longevity risk because the state bears it for them. Increasing longevity substantially raises the cost of providing for pensions, and increasing lifespans are one reason behind the underfunding of Rhode Island's pension plan. Longevity risk also increases during recessions—people tend to die later during recessions—probably because the unemployed and underemployed have more time, exercise more, and spend more time with their families during recessions.[49] Defined benefit plan sponsors then face a double whammy during recessions as asset values fall and liabilities swell as a result of increased longevity and lower interest rates.

While annuities are in theory a great deal, they are not very popular, with fewer than 6% of households buying them.[50] This is termed the *annuitization puzzle* by Modigliani (1986). Why are people reluctant to buy annuities?[51]

Credit Risk

Once you buy an annuity, you are exposed to the credit risk of the insurance company that sold you the annuity. Unfortunately for consumers, many insurance companies are no longer staid and boring. In 2011, Liberty Life Insurance Co. was acquired by Athene Holding Ltd, a company funded by the private equity firm Apollo Global Management. Before its acquisition by Athene, Liberty Life held a "squeaky clean" portfolio of state government (muni) bonds and bonds issued by highly rated corporations. After its acquisition, Liberty Life's products were backed by securitized subprime mortgage products, time-share vacation homes, and a railroad in Kazakhstan. "All the upside would go to Athene if it worked out. And the downside would go to the annuity holders if it didn't," says a former bond

[49] Stevens et al. (2012) identify one more reason: in expansions it is easier for low-skilled health-care workers (especially those working at nursing homes and outpatient centers) to find other employment. During recessions, these workers return to their low-paid jobs in health care, and thus the standard of care rises during recessions.

[50] See Inkmann, Lopes, and Michaelides (2010). And those that buy annuities often do so non-optimally. Many purchase "period certain" annuities, which are the same as bonds as they pay a certain amount every year for a fixed number of years. Period certain annuities are not the life annuities that we consider in this section. Investors would be better off buying a regular bond and simultaneously buying a deferred annuity that starts making payments when the bond matures. See Scott, Watson, and Hu (2011).

[51] All of these considerations are shared by long-term care insurance products, except for some time many insurers actually underpriced long-term care insurance. See Brown and Finkelstein (2011). One important difference between annuity markets and long-term care insurance is that the presence of Social Security's means tests serves as an impediment (it is an implicit tax) on the private market providing long-term care insurance. See Pauly (1990) and Brown and Finkelstein (2007).